

ROBOFEST 5.0

COMPLETE SPECIFICATION FILING

(Prepared by Patent Information Centre Team, GUJCOST)

Guidance for Students & Researchers

Preparation of COMPLETE SPECIFICATION (Form–2)

Autonomous Robot / Drone Inventions

(Applicable to students who have already filed a Provisional Patent Application)

1. Title of the Invention

The title shall remain exactly the same as mentioned in the provisional application

No modification or change in the title is permitted at the Complete Specification stage

2. Field of the Invention

Briefly indicate the technical field to which the invention belongs.

Example:

The present invention relates to autonomous robotic systems, and more particularly to an autonomous robot/drone configured to perform tasks using sensor-based decision making.

3. Background of the Invention

3.1 Existing Technology

Briefly describe:

Current robots/drones used for similar purposes

Their dependence on manual control or limited autonomy

3.2 Limitations of Existing Systems

- Mention technical limitations such as:
- Limited obstacle detection or navigation capability
- Inadequate real-time decision making
- Safety, reliability or power efficiency issues
- Avoid referring to specific commercial products or companies.

4. Objects of the Invention

- Clearly list the technical objectives, such as:
- To provide an autonomous robot/drone capable of real-time environment sensing
- To enable autonomous navigation and task execution

- To improve operational safety and efficiency

To reduce dependence on human intervention

5. Summary of the Invention

- Provide a concise technical overview describing:
- The overall system
- The basic autonomous working principle
- The technical contribution of the invention
- Use standard patent expressions such as:
- “The invention comprises...”
- “The system is configured to...”

6. Brief Description of Drawings

If drawings are included, list them clearly:

Fig. 1: Block diagram of the autonomous system

Fig. 2: Flow diagram of autonomous operation

7. Detailed Description of the Invention

7.1 System Architecture

- Describe the main components and their functions:
- Sensor unit for environment sensing
- Processing unit for data analysis and decision making
- Actuation unit for motion control
- Communication unit for data exchange
- Power management unit
- Descriptions should be functional and generic, not component-specific.

7.2 Autonomous Working Logic

- Explain the autonomous operation step-wise:
- Acquisition of sensor data
- Processing and interpretation of data
- Generation of control decisions
- Execution of movement or task
- Continuous feedback and correction
- Software source code should not be included.

7.3 Operational Method

Describe the overall operational flow:

- System initialization
- Autonomous task execution
- Safety handling or fault response

7.4 Alternative Embodiments

- Mention possible variations, such as:
- Ground-based robot or aerial drone
- Indoor or outdoor deployment
- Different sensor or controller configurations
- Single unit or multiple units operating together
- Conceptual embodiments are acceptable.

8. Advantages of the Invention

- List technical advantages, for example:
- Improved autonomous navigation
- Enhanced safety through real-time sensing
- Reduced human intervention
- Efficient power utilization
- Adaptability to dynamic environments

9. Industrial Applicability

- State potential applications such as:
- Agriculture
- Surveillance and monitoring
- Industrial inspection
- Disaster management
- Logistics and healthcare support

10. Drafting Guidelines for Students

- ✓ Use clear and consistent technical terms
- ✓ Explain functionality rather than performance claims
- ✓ Avoid exaggerated or promotional language
- ✓ Write in third person and present tense

11. 11. Final Student Checklist (Before Filing)

- ☐ Title matches the provisional application exactly
- ☐ Field and background clearly defined
- ☐ Technical problem clearly stated
- ☐ System components adequately described
- ☐ Autonomous working explained step-wise
- ☐ Alternative embodiments included

- ☐ Technical advantages clearly listed
- ☐ Drawings properly referenced (if any)
- ☐ No software code or unnecessary experimental data included